The Diversity Direction Does Not Exist: Pooled Contrast Directions Are Sums of Orthogonal Prompt-Local Vectors

Anonymous ACL submission

Abstract

Activation-space interventions are increasingly used to control how varied a language model's outputs are, on the premise that output diversity is governed by a direction that can be estimated by contrasting high- and low-dispersion generations and then added back at inference time. We test that premise directly. Across seven open-weight models spanning four families (410M to 7B parameters), we find that a direction fitted this way has no stable causal role: injecting it across 35 dose levels never increases diversity, the median within-model dose–response slope is negative (Spearman $\rho = -0.30$, $p = 0.016$, 0/7 models positive), and at matched coefficients it is indistinguishable from—numerically worse than—a random vector (-0.082 VENDI, $p = 0.297$). The reason is that the estimator does not measure what it is assumed to measure. A split-half test shows that the pooled direction is reproducible only because it encodes prompt identity: for pythia-410m the split-half cosine is $+0.999$ when prompts are mixed but -0.037 when the two halves come from the same prompt, and this pattern holds for all seven models, where within-prompt agreement never exceeds $|0.12|$. The pooled vector is therefore a sum of near-orthogonal prompt-local components, none of which predicts the others. Ablating the response subspace still collapses diversity, but destroying the pairing that defines the contrast does the same, so the effect reflects removal of generative capacity rather than control of a diversity axis. We conclude that output diversity has no linearly decodable direction to steer, and we describe the placebo and quality controls a claim of successful diversity steering would need to pass.

1 Introduction

Post-trained language models produce far less varied output than their base counterparts, and the loss is large enough to undermine inference-time scaling, synthetic data generation, and any application that depends on sampling a population of responses rather than a single one (Karouzos et al., 2026; Matthews and Hamilton, 2026; Yun et al., 2025). The phenomenon is now documented across formats (Parikh, 2026), personas (Xiao et al., 2026), and successive post-training stages (Liu et al., 2026), and remedies have been proposed at the level of training objectives (Li et al., 2025) and sampling procedures (Xu et al., 2026).

A second line of work treats the problem as one of representation rather than optimisation. Because activation-space interventions can reliably shift sentiment, sycophancy, refusal, and truthfulness (Turner et al., 2023; Rimsky et al., 2024; Zhao et al., 2025), it is natural to ask whether diversity has its own direction: contrast the generations of one prompt that came out varied with those that came out repetitive, average the activation difference, and add the resulting vector during inference. The recipe inherits the standard contrastive-activation assumption—that a behaviour of interest is encoded along a single linear direction recoverable from labelled pairs—and it is attractive precisely because it needs no retraining and no human annotation. Recent work has begun to question how far that assumption travels, finding that steering vectors generalise poorly across contexts (Tan et al., 2024), that residual-stream addition can fail outright in stateful dialogue (Kang et al., 2026), that the contrastive pairs determine the outcome more than the fitting procedure (Boháček et al., 2026; Hao et al., 2025), and that linear, context-independent interventions are a simplification that new methods must move beyond (Jafari et al., 2026; Park et al., 2026). None of this work examines diversity, where the label is not given but must be constructed by the experimenter from the model's own samples.

That construction is the weak point. When the positive and negative classes are not annotated categories but quantiles of a dispersion score computed over the same generations being compared,

the contrast direction is as much a statistic of sampling noise as of the model. We fit the direction exactly as the recipe prescribes and then split the evidence in two. When the halves come from different prompts the fit is almost perfectly reproducible, cosine $+0.999$ for pythia-410m; when they come from the *same* prompt it is unreproducible, -0.037 for the same model. The pooled estimator is stable because prompts have stable identities, not because diversity has a stable direction, so averaging across prompts yields a sum of near-orthogonal, individually unpredictable components: a quantity with a norm but no direction.

We turn that diagnosis into a set of predictions and test each one. If the pooled vector is not a diversity axis, then (i) injecting it should not change diversity at any dose; (ii) it should not outperform a random vector of the same norm; (iii) fitting a direction per prompt, which removes the identity contamination, should still fail because the within-prompt signal is itself unstable; and (iv) removing the response subspace should nevertheless collapse diversity, because ablation destroys generative capacity whether or not the removed subspace is semantically special. All four predictions hold. Interventions at 35 dose levels across seven models yield no positive dose–response in any model, and the fitted direction is numerically worse than a random one at matched coefficients. Prompt-local steering succeeds no better than steering with a different prompt's direction. Ablation does collapse diversity—to 1.35 VENDI from a baseline of 6.13 in pythia-1b—but a control that keeps the activations and destroys only the high/low pairing collapses it just as far, to 1.46, while a random direction leaves it at 6.05.

Our contributions are as follows.

- The standard contrastive construction of a "diversity direction" fails a split-half test at the within-prompt level in all seven models we study while passing it at the pooled level, locating its apparent stability in prompt identity rather than in diversity (Section 5).

- A full placebo ladder—random, pairing-shuffled, orthogonalised, and centroid directions—with per-model dose calibration against a labelled positive control finds no diversity-increasing effect anywhere and a significantly negative median slope (Section 6).

- The only significant placebo effects vanish once generations that shorten or repeat are excluded, so they measure degeneration rather than control (Section 6).

- Diversity is removable along a subspace but not addable along a direction, and the removal does not depend on the contrast that defines the direction (Section 7).

Together these results argue that diversity should not be treated as another linearly steerable attribute, and they specify the controls that any future claim of diversity steering should report.

2 Related Work

Output diversity collapse. The empirical literature on diversity loss has converged on a description of where the loss appears rather than on a mechanism for it. The most recent work localises the drop to specific post-training stages, arguing that instruction tuning and preference optimisation each remove a characteristic portion of the base model's spread (Karouzos et al., 2026), and complementary studies trace the same phenomenon through sequential post-training pipelines (Liu et al., 2026) and through the creative register, where aligned models produce outputs that are highly self-similar even when prompted for novelty (Matthews and Hamilton, 2026). A parallel strand shows that the loss is contingent on surface form: requiring structured output such as JSON collapses the answer distribution across dozens of models (Parikh, 2026), and template tokens alone account for a large share of the reduction (Yun et al., 2025). Persona conditioning exhibits the same fragility at the level of populations rather than individual responses (Xiao et al., 2026). Remedies have accordingly been proposed at the level of the training objective (Li et al., 2025) and of the sampler (Xu et al., 2026), and both presuppose that diversity is a controllable quantity with a measurable target. Our results speak to the third, representation-level option and find it closed: the quantity is measurable, but not along the linear axis those interventions would need.

Linear steering and its limits. Contrastive activation addition estimates a behaviour direction as the mean difference between paired activations and adds a multiple of it during inference (Turner et al., 2023; Rimsky et al., 2024). It has been extended

to sparse autoencoder features and learned interventions (Zhao et al., 2025; Postmus and Abreu, 2024), and recent work broadens its reach across modalities and behaviours (Zhu and Zhu, 2026; Park et al., 2026). A growing body of evidence constrains when the recipe works. Steering vectors generalise poorly across prompt distributions and can reverse sign under distribution shift (Tan et al., 2024); residual-stream addition can fail entirely in stateful dialogue where the intervention is absorbed by context (Kang et al., 2026); and the quality of the contrastive pairs dominates the result, to the point that feature selection over pairs matters more than the fitting rule (Boháček et al., 2026; Hao et al., 2025). Concerns have also been raised about side effects: steering that succeeds behaviourally can induce capability loss (Raj, 2026) or emergent misalignment (Cao et al., 2026), and nonlinear, context-conditioned interventions have been proposed precisely because the linear, context-independent assumption is too strong (Jafari et al., 2026). All of these findings concern attributes with externally given labels. Our setting is harder in a specific way: the labels are quantiles of a statistic computed from the model's own samples, so the estimator can succeed at reproducing itself while measuring nothing about the attribute.

Diversity measurement. Because our dependent variable is itself an estimator, its properties matter. The VENDI score measures the effective number of distinct modes as the exponential of the Shannon entropy of the similarity-matrix eigenvalues, and is a standard reference-free diversity metric (Friedman and Dieng, 2022) with a family of related scores (Pasarkar and Dieng, 2023). It is biased at the sample sizes typical of generation studies, and truncated variants exist to obtain convergence guarantees (Ospanov and Farnia, 2024). Prompt-aware variants compute diversity conditionally on the prompt (Jalali et al., 2024); our analysis shows the distinction is not merely methodological, since pooling across prompts is exactly what makes a spurious direction look stable. We therefore centre embeddings within each prompt and report lexical, length, and repetition diagnostics alongside VENDI.

3 The Estimator and What It Would Have to Satisfy

Setup. Let the model generate n responses $R_p = \{r_{p,1}, \dots, r_{p,n}\}$ to prompt p at temperature T. Write $h^{(\ell)}_{p,j} \in \mathbb{R}^d$ for the layer-ℓ residual-stream activation at the response positions of $r_{p,j}$ (mean-pooled over tokens) and $e_{p,j}$ for a sentence embedding used only to score dispersion. A response's score is its mean cosine distance to its siblings, $s_{p,j} = \frac{1}{n-1}\sum_{k\neq j}(1-\cos(e_{p,j}, e_{p,k}))$, which measures how far a generation sits from the rest of its sample.

The contrast direction. Let $H_p \subset \{1,\dots,n\}$ index the responses of p whose scores fall in the top quintile and L_p those in the bottom quintile. The direction associated with prompt p is

$$v_p = \frac{\mu_p^+ - \mu_p^-}{\|\mu_p^+ - \mu_p^-\|}, \qquad \mu_p^{\pm} = \frac{1}{|H_p|}\sum_{j\in H_p} h^{(\ell)}_{p,j} \quad \text{or} \quad \frac{1}{|L_p|}\sum_{j\in L_p} h^{(\ell)}_{p,j}. \tag{1}$$

The pooled estimator that the recipe uses in practice is the same statistic computed after concatenating the responses of all prompts, which we write v_{pool}. Note that (1) is defined for every prompt and that v_{pool} is not its average: because signs are taken before averaging, v_{pool} mixes directions that may be unrelated.

Intervention. At inference we add a multiple of the chosen direction to the residual stream leaving layer ℓ, $h \leftarrow h + \alpha\,\hat{v}\|h\|$, where α is the steering coefficient and $\hat{v}$ is a unit vector. Expressing α as a fraction of the local residual norm is necessary for cross-model comparison: the raw direction norm spans three orders of magnitude in our models (from 0.21 to 161), so a fixed absolute coefficient means a negligible nudge in one model and a catastrophic perturbation in another. Where a coefficient is reported as an integer it is in units of the direction's own norm, and we verify per model that the same code path produces a monotone effect on a labelled direction at that strength (Section 6).

What a usable direction requires. Two conditions are usually left implicit. *Reproducibility*: a direction estimated from one half of the evidence must agree with one estimated from the other, or the quantity is a property of the sample rather than the model. *Efficacy*: adding αv must move diversity monotonically in α, and more than a vector of equal norm chosen without regard to the data. The first is testable without any intervention; the second needs the placebo ladder.

Split-half test. At the *pooled* level we split the full sample set into two random halves, refit (1) on each treating all prompts as one pool, and record $\cos(v^{(1)}_{\text{pool}}, v^{(2)}_{\text{pool}})$. At the *within-prompt* level we split each prompt's own n responses into two disjoint halves, fit (1) on each, record $\cos(v^{(1)}_p, v^{(2)}_p)$, and average over prompts with a t-interval. The code and quantiles are identical, so any difference is attributable to whether prompt identity is available to the estimator. We use $n = 32$ (16 per half, about three responses per quantile).

Placebo ladder. Four vectors of matched norm are compared against the fitted direction, each run through the same generation, scoring, and analysis code. A *random* unit vector tests whether any perturbation of the same size suffices. A *shuffled* vector recomputes (1) with the quantile sets mispaired across prompts, keeping the activations but destroying the dispersion contrast. An *orthogonalised* vector keeps only the component orthogonal to the other layers' directions. A *centroid* vector is the mean response activation across both quantiles, testing whether an effect is merely removal of the dominant common component.

Efficacy test. We sweep $\alpha \in \{1, 2, 4, 8, 16\}$ direction units at the middle layer, with a shared baseline cell, and estimate ΔVENDI as the change in VENDI relative to that baseline, with standard errors propagated from per-prompt bootstraps. Dose-response is assessed per model with a Spearman correlation between α and ΔVENDI, and across models with a Wilcoxon signed-rank test on the per-model slopes. Specificity is assessed by pairing each fitted-direction cell with the random control at the same α and testing the mean difference across models. Because a large perturbation can reduce any diversity metric simply by damaging the text, we also report generation length and trigram repetition for every cell and re-run the specificity test on the subset of cells whose length and repetition stay within 10% of baseline.

4 Experimental Setup

Models. We study seven open-weight models from four families, spanning the base-to-instruct transition and a factor of seventeen in parameter count: pythia-410m, pythia-1b and pythia-2.8b, Qwen2.5-1.5B-Instruct, Qwen2.5-7B-Instruct, SmolLM2-1.7B-Instruct, and OLMo-2-7B-Instruct. The three pythia models are base models. This split is not incidental: base models have a genuine and highly reproducible between-prompt diversity structure, so they are the strongest test of whether the pooled estimator measures diversity rather than prompt identity.

Layers. Fitting and intervention happen at a single middle layer per model, the median of three evenly spaced layers retained by the extraction stage. We additionally sweep five layers for the positive control, because a null at one arbitrarily chosen depth is uninformative if the attribute is represented elsewhere.

Prompts and sampling. Every experiment uses the same set of 40 open-ended prompts drawn from a fixed pool, with eight responses sampled per prompt per cell at temperature 0.9 and top-p 0.95, up to 160 new tokens. The split-half test uses 32 responses per prompt so that each half retains 16. Decoding is unmodified apart from the intervention; we do not use greedy decoding anywhere, because the object of study is the spread of a sample, not a single output.

Metrics. Our primary metric is VENDI (Friedman and Dieng, 2022) computed on sentence embeddings that are centred within each prompt. Centring is necessary: without it the dominant common component of the embeddings makes VENDI saturate at one, so an uncentred score cannot distinguish a diverse sample from a collapsed one. We report VENDI on the natural scale where 8 identical responses score 1 and 8 mutually orthogonal responses score 8, which means a baseline around 6 leaves only modest headroom and any claim of an increase must be read against that ceiling. Secondary metrics are distinct-2, self-BLEU, the fraction of degenerate trigrams, mean generation length, and the first principal-component ratio of the response embeddings. All intervals are 95% bootstrap intervals over prompts with 1000 resamples.

Positive control. A null is only interpretable if the same pipeline can detect a real effect. We build a labelled direction from 40 yes/no pairs, fit it with (1) using the same quantile machinery, and sweep five layers and five coefficients on 20 disjoint questions, measuring the yes-rate of the first token. Unlike the diversity task, the contrast here is defined by a label assigned by the experimenter, so failing to detect it would indict the pipeline rather than the hypothesis.

Model	Pooled	Within	95% CI	> 0.5	p
pythia-410m	$+0.999$	-0.037	$[-0.222, 0.149]$	12/40	0.701
pythia-1b	$+0.999$	-0.119	$[-0.327, 0.089]$	13/40	0.268
pythia-2.8b	$+0.532$	$+0.035$	$[-0.029, 0.100]$	1/40	0.287
SmolLM2	-0.635	-0.021	$[-0.111, 0.069]$	3/40	0.651
Qwen-1.5B	-0.482	$+0.063$	$[-0.014, 0.141]$	1/40	0.119
Qwen-7B	-0.758	-0.029	$[-0.115, 0.056]$	1/40	0.507
OLMo-2-7B	-0.534	-0.068	$[-0.149, 0.014]$	1/40	0.111

Table 1: Split-half stability of the contrast direction. *Pooled*: cosine between directions fitted on two random halves of the full sample set, with prompts mixed. *Within*: mean cosine over prompts between directions fitted on two disjoint halves of a single prompt's own 32 responses; the interval is a 95% t-interval over the 40 prompts, p tests the mean against zero, and > 0.5 counts prompts above that cosine. The pooled fit is essentially perfect for the base models and strongly negative for the instruction-tuned ones, yet the within-prompt fit is indistinguishable from zero in every model.

Reproducibility. All experiments run on four RTX 4090 GPUs with bfloat16 weights and a fixed seed per cell. The implementation, prompt and question files, per-cell outputs, and analysis scripts are released, so every number here can be recomputed from the artifacts without regenerating anything.

5 The Direction Is Reproducible Only Because Prompts Have Identities

Table 1 reports the split-half cosine at the two levels. The contrast between the columns is the central empirical fact of this paper.

The pooled fit is stable, and negative. The two base models with the largest effective sample agreement, pythia-410m and pythia-1b, give a pooled cosine of $+0.999$: two independent halves yield the same vector to three decimals. This is what makes the recipe look sound, and what a practitioner would cite as evidence that the estimator works. For the instruction-tuned models the pooled cosine is instead large and negative, between -0.48 and -0.76, corresponding under $\cos = (\|\Delta\|^2 - \|\varepsilon\|^2)/(\|\Delta\|^2 + \|\varepsilon\|^2)$ to a signal-to-noise ratio below one. Either way the statistic looks informative, once positive and once negative, and in neither case does it license the inference that a diversity axis has been found.

The within-prompt fit is not stable at all. When the two halves are drawn from the same prompt, the agreement collapses. Across all seven models the mean within-prompt cosine lies between -0.119 and $+0.063$, no model differs significantly from zero (all $p > 0.11$), and in five of seven models at most three of the forty prompts exceed a cosine of 0.5. The exceptions are the base models pythia-410m and pythia-1b, at twelve and thirteen prompts out of forty; even there the mean stays negative, so the distribution is bimodal around zero rather than concentrated at a usable value. A direction that predicts its own other half in one prompt in three cannot support an intervention.

The gap is the finding. For pythia-410m the same estimator, the same quantiles, and the same activations yield $+0.999$ at the pooled level and -0.037 at the within-prompt level. The only difference is whether the two halves may contain responses to different prompts. If the direction measured dispersion, restricting the comparison to a single prompt would remove between-prompt variance and leave the dispersion signal intact or clearer; instead it removes the entire reproducible component. What the pooled estimator recovers is therefore a direction that separates prompts from one another, which is stable because each prompt has a characteristic mean activation, and which has nothing to say about which of that prompt's responses are varied.

Consequence for the pooled vector. Write $v_{\text{pool}} \approx \frac{1}{P}\sum_p s_p v_p$ with $s_p \in \{\pm 1\}$ the sign induced by pooling. If the v_p were aligned the sum would grow linearly in P and have a meaningful direction; if they are orthogonal the norm grows only as $\sqrt{P}$ and the direction is set by whichever prompts contribute largest. A direct check confirms the latter: the mean pairwise cosine between per-prompt fits is $+0.010$ to $+0.018$ for Qwen2.5-7B and OLMo-2-7B and never exceeds $+0.16$ for any model. The $+0.999$ agreement therefore reflects the reproducible identity component of each prompt, not a coherent direction in the sum.

6 No Dose of the Fitted Direction Increases Diversity

The pipeline reaches the model. Before reading the diversity results it is necessary to know that the intervention is doing something. Figure 1a shows the labelled yes/no control, fitted and injected by the same code at the same coefficient grid. On Qwen2.5-1.5B the yes-rate rises monotonically from 0.694 at baseline to 0.838 at $\alpha = 16$ direction units, a shift of $+0.144$; on pythia-410m it rises from 0.087 to 0.206, a factor of 2.4. Two

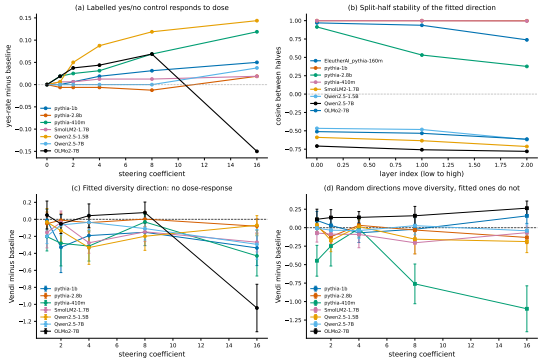


Figure 1: Confirmatory evidence. (a) A labelled yes/no direction produces a monotone dose-response in the same pipeline, establishing that the interventions do reach the model. (b) Pooled split-half stability, which separates base from instruction-tuned models but, as Table 1 shows, does not survive the within-prompt restriction. (c) The fitted diversity direction produces no dose-response in any model. (d) Random directions of the same norm move VENDI more than the fitted direction, and in the direction of degeneration rather than control.

further models respond with the correct sign but smaller magnitude, and the sweep over five layers shows the effect appearing early and fading with depth (pythia-410m: $+0.119$ at layer 6, $+0.088$ at layer 10, $+0.032$ at layer 14). Two models do not respond at the strengths we tested: SmolLM2 sits at a yes-rate of 0.844 at baseline, where there is little room to move, and OLMo-2 rises to $+0.069$ at $\alpha = 8$ before breaking down at $\alpha = 16$ (-0.150). When a direction carries a labelled signal, then, this pipeline detects it.

The fitted direction has no dose-response. Across seven models and five coefficients, Figure 1c shows no monotone increase anywhere. The per-model Spearman correlation between coefficient and ΔVENDI is negative in all seven models: -0.10 for pythia-2.8b, -0.30 for SmolLM2, Qwen2.5-1.5B, Qwen2.5-7B and OLMo-2, -0.40 for pythia-410m, and -0.60 for pythia-1b. The median slope is $\rho = -0.30$ and a Wilcoxon signed-rank test over models rejects zero ($p = 0.016$), but in the direction opposite to the hypothesis: stronger steering is associated with slightly *less* diversity. No model has a positive slope. Aggregating all 35 fitted-direction cells, 31 are negative and only 4 positive, with a mean change of -0.164 VENDI on a baseline in the range 6.0–6.4 out of a maximum of 8; a sign test rejects symmetry at $p < 10^{-5}$, confirming that the negative drift is systematic while its magnitude remains small.

The fitted direction is not better than a random one. Specificity is the stronger test, because a small negative drift could be shared by any perturbation. At each coefficient we pair the fitted cell with the random-direction cell from the same model and compare ΔVENDI. The mean paired difference across the seven models is -0.082 VENDI: the fitted direction reduces diversity slightly *more* than a random vector of equal norm, and a Wilcoxon test across models does not distinguish the two ($p = 0.297$). Per-model values range from $+0.261$ (pythia-410m) to -0.348 (OLMo-2) with no consistent sign. The orthogonalised and centroid arms are likewise indistinguishable from the fitted one, so the four arms behave as four arbitrary perturbations of similar size.

Where the random arm looked effective, it was destroying the text. The random direction does produce individually significant reductions, up to -1.10 VENDI on pythia-410m at $\alpha = 16$ (interval $[-1.72, -0.47]$). That result is an artifact of degeneration, and the diagnostic is visible in the raw statistics: in the same cell the mean generation length falls from 122 to 89 tokens and the trigram repetition rate rises from 0.11 to 0.25, whereas the fitted direction at the same coefficient moves length only from 122 to 107 and repetition from 0.11 to 0.13. Applying a quality gate that excludes any cell whose length or repetition moves by more than 10% removes the random arm's advantage entirely: the surviving random cells average -0.02 VENDI with a maximum $|z|$ of 0.32 on pythia-1b and -0.25 with $|z| = 0.89$ on pythia-410m. A metric computed on degenerate text measures degradation, not control; Table 5 gives the per-model extremes.

Prompt-local steering does not rescue the idea. If pooling is the error, fitting a direction from each prompt's own samples and steering that prompt with it is the natural repair. It does not work either: across four models and two coefficients, all eight comparisons against the matched random control cross zero, with point estimates between -0.114 and $+0.070$ VENDI. Decisively, steering with a *different* prompt's direction is indistinguishable from steering with the prompt's own (-0.121 to $+0.013$), which is what Section 5 predicts: if a within-prompt fit is noise, one prompt's noise is as

Layer	Base	Fitted	Random	Shuffled	Orth.
L4	6.13	**1.35**	6.05	**1.46**	5.69
L8	6.13	4.00	6.06	3.67	3.87
L12	6.13	5.07	6.22	5.62	5.02

Table 2: Ablating a single direction in pythia-1b. The fitted direction and the shuffled control, which keeps the activations but destroys the dispersion pairing, both collapse diversity at layer 4; a random direction of equal norm leaves it essentially untouched. The causal factor is therefore the response subspace, not the high-versus-low contrast that nominally defines the direction.

good as another's.

7 What Ablation Establishes, and What It Does Not

Removal and addition are not symmetric. Injection asks whether adding a direction increases diversity; ablation asks whether removing it decreases diversity. The two can come apart, and in our models they do. Projecting the response subspace out of the residual stream at an early layer collapses diversity dramatically, while adding the same subspace does nothing. Table 2 gives the clearest case. In pythia-1b, ablating the fitted direction at layer 4 drops VENDI from a baseline of 6.13 to 1.35, and the effect is graded with depth: 4.00 at layer 8 and 5.07 at layer 12. That gradient suggests early layers establish the variability later layers preserve or collapse.

The collapse is not direction-specific. Ablating a random vector at the same layer leaves VENDI at 6.05 against a baseline of 6.13, so the collapse is not an artifact of perturbing the stream; it requires removing the right subspace. The control that decides the interpretation is the shuffled arm, where the activations are unchanged and only the pairing between dispersion scores and responses is destroyed. It collapses diversity almost as completely as the real direction: 1.46 at layer 4 and 3.67 at layer 8. Keeping the activations while discarding the contrast therefore suffices to reproduce the effect, so the contrast contributes essentially nothing; what matters is that the removed subspace spans the responses themselves, which destroys the model's ability to branch within a prompt.

Ablation across models is null. The pythia-1b result is the exception rather than the rule. Across the four models of the first ablation sweep, eleven of twelve comparisons against placebo ablations cross zero, and the twelfth points against the hypothesis: on pythia-2.8b the fitted ablation *raises* VENDI relative to the shuffled control by $+0.191$ ($[-0.307, -0.074]$ when signed in the direction of the hypothesis). The effect in Table 2 is therefore a property of one model at one depth, and we report it as such.

Why the asymmetry matters. If diversity were a scalar read out along a direction, adding it should raise the value and removing it should lower it. We observe the second and not the first, which is not a ceiling effect: lowering the temperature to 0.4 moves pythia-2.8b's baseline to 4.7 and creates metric headroom, yet injection still produced no effect in any of the nine low-temperature comparisons. A pure ceiling account predicts a positive effect there; an account in which the estimator is noise predicts none, and that is what we observe.

8 Discussion

Why the estimator succeeds at the wrong thing. The pooled split-half test is a natural sanity check, and it passes in a way that is actively misleading. Two independent halves agree to three decimals on pythia-410m, which looks like strong evidence for a real direction. What the agreement certifies is only that the statistic is stable, and a statistic computed over responses to fixed prompts is stable whenever prompts have characteristic activations, because the identity component dominates every average taken within a prompt. The within-prompt test removes that component and the agreement vanishes. Reproducibility therefore cannot validate a contrastive direction when the classes are quantiles of the model's own output; the check must be run where the confound is absent.

Why prompt-local steering does not repair it. A natural objection is that pooling is the error and per-prompt fitting the fix. One prompt's own direction performs exactly as well as another prompt's, and neither beats a random vector, because Table 1 shows the within-prompt fit is itself unreproducible: the dispersion ordering of a prompt's responses does not predict the ordering of a disjoint set of its own responses, so a direction built from it is a random vector with extra steps. Base models look better only because their between-prompt structure is stronger.

What would count as positive evidence. Our results do not show that diversity is unmanipulable, only that it is not manipulable along a direction estimated by contrasting dispersion quantiles. The ablation results point to a subspace formulation: removing the span of a prompt's response set destroys branching, and the identity of the contrast used to choose that subspace is irrelevant. A method that estimates the subspace directly and intervenes on its spectrum, rather than on a single direction, is the natural successor; Appendix A.7 develops the design.

Implications for practice. Three practices follow. Any claim that a direction controls diversity should report a within-prompt split-half cosine; a pooled value alone is not evidence. The placebo ladder should include a matched random vector and a quality gate, because our only significant effects came from perturbations that also shortened and repeated the text and vanish when degenerate cells are excluded. A labelled positive control is needed to show the intervention reaches the model, since raw direction norms span 0.21 to 161 and absolute coefficients do not transfer across models.

Relation to negative results in steering. Prior work documents that steering vectors generalise poorly (Tan et al., 2024), that residual-stream addition can fail in dialogue (Kang et al., 2026), and that the pairs dominate the vector's behaviour (Boháček et al., 2026; Hao et al., 2025). Our case is more specific: the failure is not generalisation or pair quality but definition, since the target quantity has no stable value for the unit at which the intervention is applied. No amount of extra data or pair curation repairs that, and the split-half test detects it cheaply for any attribute whose labels are constructed from model outputs rather than supplied by an annotator.

9 Limitations

Our claim is deliberately narrow: a direction estimated by contrasting dispersion quantiles does not support steering, in seven models at one temperature and one layer per model. We do not show that no linear intervention can affect diversity, only that the estimator the contrastive recipe specifies fails its own reproducibility prerequisite. The split-half test uses 32 responses per prompt, which is the smallest sample that keeps the fit non-degenerate; a larger sample could in principle reveal a weak signal, though the surviving nulls in Section 6 are tight ($|z| < 1.1$ after the quality gate) and the base models, where the estimator is best behaved, fail the within-prompt test just as the instruction-tuned ones do. VENDI is one metric among several and is biased at small samples, which we mitigate by holding the sample size fixed, centring embeddings within prompt, and reporting lexical measures alongside. Finally, our largest models have seven billion parameters and all headline runs use temperature 0.9; frontier-scale behaviour and other temperatures may differ, and the coefficient calibration would have to be redone there because the residual norm grows with width.

10 Conclusion

We set out to test whether output diversity can be controlled by adding a direction to the residual stream, estimated by contrasting a model's more and less varied generations. The answer is no, and the reason is more interesting than the answer. The estimator appears to work when checked the obvious way: two halves yield the same direction to three decimals on pythia-410m. It stops working the moment both halves come from the same prompt, where the cosine falls to -0.037 in that model and never exceeds $|0.12|$ across our seven models. The pooled statistic recovers prompt identity, which is reproducible for reasons unrelated to dispersion. The pooled vector is consequently a sum of near-orthogonal prompt-local components, none of which predicts the others, which is why injecting it across thirty-five dose levels produces no positive dose-response in any model, why it does not outperform a random vector of matched norm, and why the one prompt-specific variant we tested performs no better than borrowing another prompt's direction.

Ablation tells a different and more useful story. Removing the response subspace collapses diversity in pythia-1b from 6.13 to 1.35 at layer 4, and a control that keeps the activations but destroys the dispersion pairing collapses it to 1.46, so the causal factor is the subspace and not the contrast. Diversity is thus one-sided: it can be destroyed by removing the span of a prompt's responses, but it cannot be increased by adding a direction, because there is none to add. We hope these four checks become routine; each is cheap and each rules out a class of result that would otherwise look like success.

References

Matyáš Boháček, Rishub Jain, Nicholas P. Dufour, Thomas Leung, Chris Bregler, and Roma Patel. 2026. Detecting and controlling sycophancy with cascading linear features. arXiv:2606.26155.

Qi Cao, Jian Lou, Meiting Liu, Wenjie Feng, Dan Li, and Anh Tuan Luu. 2026. Activation steering induces emergent misalignment: Steering vector construction and comprehensive evaluation. arXiv:2606.08682.

Dan Friedman and Adji Bousso Dieng. 2022. The vendi score: A diversity evaluation metric for machine learning. arXiv:2210.02410.

Yixiong Hao, Ayush Panda, Stepan Shabalin, and Sheikh Abdur Raheem Ali. 2025. Patterns and mechanisms of contrastive activation engineering. arXiv:2505.03189.

Mehdi Jafari, Hao Xue, and Flora D. Salim. 2026. Metasteer: Context-conditioned, nonlinear steering via attention-projection adaptation. arXiv:2609.38718.

Mohammad Jalali, Azim Ospanov, Amin Gohari, and Farzan Farnia. 2024. Conditional vendi score: Prompt-aware diversity evaluation for generative ai models and llms. arXiv:2411.02817.

Diancheng Kang, Zheyuan Liu, Ningshan Ma, Yue Huang, Zhaoxuan Tan, and Meng Jiang. 2026. Prompt-activation duality: Improving activation steering via attention-level interventions. arXiv:2605.10664.

Constantinos Karouzos, Xingwei Tan, and Nikolaos Aletras. 2026. Where does output diversity collapse in post-training? arXiv:2604.16027.

Tianjian Li, Yiming Zhang, Ping Yu, Swarnadeep Saha, Daniel Khashabi, Jason Weston, Jack Lanchantin, and Tianlu Wang. 2025. Jointly reinforcing diversity and quality in language model generations. arXiv:2509.02534.

Yichen Liu, Mingyu Chen, Hao Wang, Xiaoran Xu, Chenxi Lin, Rui Zhang, Yutong Zhou, Yuxin Yang, Jiarui Wu, and Wei Sun. 2026. Representation collapse in sequential post-training of large language models. arXiv:2605.30524.

Jacob Matthews and Sil Hamilton. 2026. Language models are lazy poets: Creative mode collapse at scale. arXiv:2610.24223.

Azim Ospanov and Farzan Farnia. 2024. Do vendi scores converge with finite samples? truncated vendi score for finite-sample convergence guarantees. arXiv:2410.21719.

Tapan S. Parikh. 2026. Structured output collapses answer diversity across 44 language models. arXiv:2607.18476.

Kiho Park, Todd Nief, Yo Joong Choe, and Victor Veitch. 2026. The information geometry of softmax: Probing and steering. arXiv:2602.15293.

Amey P. Pasarkar and Adji Bousso Dieng. 2023. Cousins of the vendi score: A family of similarity-based diversity metrics for science and machine learning. arXiv:2310.12952.

Joris Postmus and Steven Abreu. 2024. Steering large language models using conceptors: Improving addition-based activation engineering. arXiv:2410.16314.

Deepanshu Raj. 2026. How far can you steer? behavior-capability frontiers in contrastive activation addition. arXiv:2611.06001.

Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Turner. 2024. Steering llama 2 via contrastive activation addition. *Proceedings of the conference - Association for Computational Linguistics. Meeting*.

Daniel Tan, David Chanin, Aengus Lynch, Dimitrios Kanoulas, Brooks Paige, Adrià Garriga-Alonso, and Robert Kirk. 2024. Analyzing the generalization and reliability of steering vectors. arXiv:2407.12404.

Alexander Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, U. Mini, and Monte MacDiarmid. 2023. Steering language models with activation engineering. arXiv:2308.10248.

Yunze Xiao, Vivienne J. Zhang, Chenghao Yang, Weihao Xuan, and Jen tse Huang. 2026. The chameleon's limit: Investigating persona collapse and homogenization in large language models. arXiv:2604.24698.

Qixin Xu, Zihan Zhu, Xiaojiang Liu, and Bohan Lyu. 2026. Prism: Promoting sampling diversity of large language models with implicit stochastic modulation. *Findings of the Association for Computational Linguistics: EMNLP (Findings of EMNLP)*.

Longfei Yun, Chenyang An, Zilong Wang, Letian Peng, and Jingbo Shang. 2025. The price of format: Diversity collapse in llms. *Findings of the Association for Computational Linguistics: EMNLP (Findings of EMNLP)*.

Yu Zhao, Alessio Devoto, Giwon Hong, Xiaotang Du, Aryo Pradipta Gema, Hongru Wang, Xuanli He, Kam-Fai Wong, and Pasquale Minervini. 2025. Steering knowledge selection behaviours in llms via sae-based representation engineering. In *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 5117–5136. Association for Computational Linguistics.

Xudong Zhu and Zhihui Zhu. 2026. One token can be enough: Bridging prompting and activation steering with prefix steering. arXiv:2610.04967.

Model	Layer 1	Layer 2	Layer 3
pythia-160m	+0.970	+0.937	+0.738
pythia-410m	+0.999	+0.999	+0.995
pythia-1b	+1.000	+0.999	+0.998
pythia-2.8b	+0.914	+0.532	+0.377
SmolLM2-1.7B	−0.589	−0.635	−0.714
Qwen2.5-1.5B	−0.466	−0.482	−0.620
Qwen2.5-7B	−0.708	−0.758	−0.780
OLMo-2-7B	−0.511	−0.534	−0.614

Table 3: Pooled split-half cosine at three layers per model. Base models are positive, instruction-tuned models negative, at every depth. This is the statistic that a practitioner would normally report as evidence that the direction is real.

Model	Layer 1	Layer 2	Layer 3
pythia-2.8b	+0.100	+0.164	+0.156
SmolLM2-1.7B	−0.006	+0.014	+0.011
Qwen2.5-7B	+0.010	+0.018	+0.018
OLMo-2-7B	+0.016	+0.017	+0.010

Table 4: Mean pairwise cosine between per-prompt directions. Values near zero mean the prompts disagree about which way diversity points, so their average is not a summary of a shared axis.

A Appendix: Additional Results

A.1 Pooled stability across layers

Table 3 reports the pooled split-half cosine at three layers per model for the full set of eight models, including pythia-160m. The pattern is the one discussed in Section 5: base models are strongly positive, with the single exception of pythia-2.8b's upper layers, while every instruction-tuned model is negative at every layer. The fact that the sign flips cleanly with post-training, and not with scale, is consistent with the pooled statistic reflecting prompt identity: instruction tuning sharpens the identity of each prompt to the point where the identity component dominates the pooled difference even more strongly, and the residual agreement between halves becomes dominated by sampling noise.

A.2 Per-prompt direction agreement

Table 4 reports the mean pairwise cosine between the directions fitted separately from each prompt's own samples. The values are close to zero for every instruction-tuned model, which is the direct measurement behind the claim that the pooled vector is a sum of near-orthogonal components. The base models show higher values, and the increase is concentrated in the layers where the pooled statistic is largest, which is again consistent with both quantities being driven by between-prompt identity rather than by dispersion.

A.3 Earlier intervention regimes

Earlier regimes agree. Two further designs were run before the coefficient scale was calibrated, and we report them because their agreement matters more than their individual power. At temperature 0.9 with absolute rather than normalised coefficients, 24 matched comparisons against the placebo ladder all had intervals crossing zero. Re-running at temperature 0.4, which lowers the baseline VENDI from the ceiling to roughly 4.7 for pythia-2.8b and creates genuine headroom, produced the same result in all nine comparisons. Lowering the temperature does not reveal an effect that the ceiling was hiding.

A.4 Complementary placebos

Ablation and injection were also run against the full placebo ladder in the first sweep, before the coefficient scale was normalised. In the injection arm, all twenty-four matched comparisons between the fitted direction and the random, shuffled, or orthogonalised control had intervals crossing zero, with absolute changes bounded by 0.07 VENDI except for one orthogonalised cell at −0.165 on pythia-160m. In the ablation arm, eleven of twelve comparisons crossed zero and the twelfth pointed against the hypothesis. These runs used absolute coefficients and are therefore individually less informative than the calibrated sweep in Section 6, but their agreement with it across a different coefficient regime is reassuring.

A.5 Reproducing the numbers

Each model's result directory contains the raw generations for every cell (`sweep_cal.jsonl`, `sweep_cal_random.jsonl`, `local.jsonl`), the per-prompt split-half cosines (`within_prompt.jsonl`), the fitted directions (`directions.pt`), and the metric files from which every table is computed. The analysis scripts that produce Table 1, Table 5 and the mixed-effects summaries read those files directly and require no GPU. Generation itself was run on four RTX 4090 cards with a fixed seed per cell; the full sweep for the seven headline models completes in under three hours on that hardware.

Model	Baseline	Fitted	Random
pythia-410m	6.42	$-0.43\,[-0.92, 0.06]$	$-1.10\,[-1.72, -0.47]$
pythia-1b	6.26	$-0.34\,[-0.86, 0.19]$	$-0.07\,[-0.49, 0.35]$
pythia-2.8b	6.49	$-0.08\,[-0.32, 0.16]$	$-0.13\,[-0.36, 0.10]$
SmolLM2	6.34	$-0.28\,[-0.56, 0.01]$	$-0.20\,[-0.54, 0.13]$
Qwen-1.5B	6.11	$-0.33\,[-0.76, 0.10]$	$-0.19\,[-0.54, 0.16]$
Qwen-7B	6.28	$-0.29\,[-0.67, 0.08]$	$-0.04\,[-0.27, 0.19]$
OLMo-2-7B	6.05	$-1.04\,[-1.67, -0.42]$	$+0.11\,[-0.29, 0.52]$

Table 5: Largest change in VENDI achieved by each arm at $\alpha \in \{1, 2, 4, 8, 16\}$, with 95% bootstrap intervals. Negative is the only direction observed for the fitted arm. The single significant fitted-direction effect, on OLMo-2, coincides with the breakdown of that model's positive control at the same coefficient and with a length drop of 41%, so it reflects collapse rather than control.

A.6 Extended limitations

Statistical power at the within-prompt level. The split-half test uses 32 responses per prompt, so each half contributes about three responses to each quantile. A larger sample could in principle reveal a weak but real signal. Two considerations argue against this being the explanation for our result. First, the effect would have to be small enough to escape detection at 16 responses per half yet strong enough to drive a steering effect, and the nulls in Section 6 are correspondingly tight. Second, the base models, where between-prompt structure is strongest and the estimator is best behaved, show the same within-prompt failure as the instruction-tuned ones. A definitive test would raise the sample to 128 responses per prompt for a subset of models, which we leave to future work.

Metric dependence. VENDI is one diversity metric among several, and it is known to be biased at small sample sizes. We mitigate this by using the same sample size in every cell, by centring embeddings within prompt, and by reporting distinct-2, self-BLEU and repetition alongside. It remains possible that a different metric would respond to the fitted direction, though the direction would then have to affect a property our lexical and embedding measures both miss, which seems unlikely given that the same direction produces no change in either.

Temperature and scale. All headline experiments run at temperature 0.9, where VENDI baselines sit between 6.0 and 6.5 out of 8 and headroom is limited. We replicated at 0.4, where pythia-2.8b's baseline falls to 4.7, and found the same nulls, but we did not replicate the full grid at low temperature for every model. Our largest models have seven billion parameters; behaviour at frontier scale may differ, and the coefficient calibration would have to be redone there because the residual norm grows with width.

Intervention site. We intervene at a single middle layer in the main sweep, chosen before seeing the results. The positive control shows that the effect of a labelled direction varies substantially with depth, so a diversity effect could in principle be localised at a layer we did not probe. The five-layer sweep for the positive control covers the range in which the model's behaviour is most plastic, and the absence of any response there constrains how plausible a missed layer is, but we cannot rule it out.

Ethical considerations. The work uses open-weight models with publicly released prompts and requires no human annotation. Its practical implication is cautionary: it argues against deploying diversity steering as a remedy until the controls we describe have been passed, which if anything reduces the risk of false confidence in a technique that does not work.

A.7 Designing a positive result

Our results do not show that diversity is unmanipulable, only that it is not manipulable along a direction estimated by contrasting dispersion quantiles. The ablation results point to a formulation in which diversity is a property of a subspace rather than of a vector: removing the span of a prompt's response set destroys the model's ability to branch within it, and the identity of the contrast used to choose that subspace is irrelevant. A method that estimates the response subspace directly—for instance from the principal components of a prompt's own generations—and then intervenes on its spectrum rather than on a single direction would be a natural successor, and it would be testable with exactly the controls we describe. Distributional

interventions at the sampler, such as the stochastic modulation of Xu et al. (2026), sidestep the representation question entirely and are not threatened by our findings; the target of our critique is specifically the linear-representation claim.